

BARIS ASKIN

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EDUCATION

Carnegie Mellon University, Pittsburgh, USA

Fall 2022 - Spring 2027 (Anticipated)

· Ph.D. in Electrical and Computer Engineering (M.Sc. en route, 2025)

GPA: 4.0/4.0

Advisors: Gauri Joshi and Carlee Joe-Wong

Bilkent University, Ankara, Turkey

2017 - 2022

· B.Sc. in Electrical and Electronics Engineering

GPA: 3.99/4.0

Valedictorian – Highest Ranked Graduate of 2022 in the Faculty of Engineering

RESEARCH AREAS

LLMs, Interpretability, Alignment, Post-Training & RL (RLVR, distillation, fine-tuning), LLM Routing, Distributed/Federated Learning

WORK & RESEARCH EXPERIENCE

Graduate Researcher

August 2022 – Present

Carnegie Mellon University

- LLM Alignment: emergent & subliminal misalignment analysis through training data and signal (SFT, on-/off-policy distillation) [[NeurIPS'26](#)][[Preprint](#)]
- LLM Interpretability: info.-theoretic method for internal planning horizon and branch awareness [[ICLR'26](#)]
- LLM RLVR: federated GRPO with off-policy coordination on public data [[ICML'26 Workshop](#)]
- LLM Routing: learning routers from sparse, decentralized evaluations in federated settings [[NeurIPS'26](#)]
- Federated Learning and Fine-Tuning: communication-efficient fine-tuning (LoRA) and asynchronous FL with convergence guarantees [[NeurIPS'25](#)][[AISTATS'25](#)][[UAI'24](#)]

Research Intern

May 2025 – August 2025

NVIDIA Corporation

- Addressed client staleness in large-scale async FL, where outdated updates degrade the global model
- Developed a method to better utilize stale updates via data-free knowledge distillation [[NeurIPS'26](#)]
- My method was integrated into NVIDIA FLARE, NVIDIA's open-source FL framework [[Code](#)]

Quant Researcher Intern

June 2026 – August 2026

IMC Trading

- Quantitative research on index options
- Analyzed large-scale trade data and developed statistical methods to detect patterns

SELECTED PUBLICATIONS

1. Emergent and Subliminal Misalignment Through the Lens of Data-Mediated Transfer

B. Askin*, M. Ustaomeroglu*, A. Nayak*, G. Joshi, G. Qu, C. Joe-Wong

NeurIPS 2026 [[Link](#)]

2. Internal Planning in Language Models: Characterizing Horizon and Branch Awareness

M. Ustaomeroglu*, B. Askin*, G. Joshi, C. Joe-Wong, G. Qu

*equal contribution

ICLR 2026 [[Link](#)][[Poster](#)][[Video](#)]

3. Reviving Stale Updates: Data-Free Knowledge Distillation for Asynchronous FL

B. Askin, H. R. Roth, Z. Sun, C. Joe-Wong, G. Joshi, Z. Xu

NeurIPS 2026 [[Link](#)][[Code](#)]

4. **Federate the Router: Learning LM Routers with Sparse and Decentralized Evaluations**
B. Askin*, S. Patel*, A. Nayak*, A. Vigano, J. Woo, G. Joshi, C. Joe-Wong
NeurIPS 2026 [[Link](#)][[Code](#)]
5. **PubSwap: Public-Data Off-Policy Coordination for Federated RLVR**
A. Nayak*, B. Askin*, M. Ustaomeroglu, C. Joe-Wong, G. Joshi
ICML 2026 Workshop on RL from World Feedback [[Link](#)]
6. **Federated Communication-Efficient Multi-Objective Optimization**
B. Askin, P. Sharma, G. Joshi, C. Joe-Wong
AISTATS 2025 [[Link](#)][[Code](#)][[Poster](#)]
7. **Ravan: Multi-Head Low-Rank Adaptation for Federated Fine-Tuning**
A. Raje, B. Askin, D. Jhunjhunwala, G. Joshi
NeurIPS 2025 [[Link](#)]
8. **FedAST: Federated Asynchronous Simultaneous Training**
B. Askin, P. Sharma, C. Joe-Wong, G. Joshi
UAI 2024 [[Link](#)][[Code](#)][[Poster](#)]
9. **Aligned Data Can Induce Misalignment via Context Confusion**
Y. Bakman, D. N. Yaldiz, B. Askin, S. Roy, M. Ziyadi, S. Avestimehr, S. P. Karimireddy
Preprint 2026 [[Link](#)]
10. **Federated Learning is a Lens towards a Democratized Future for the Scaling Law Era**
H. H. Jiang, B. Askin, G. Joshi, C. Joe-Wong
ICML 2026 (Position Track) [[Link](#)]

Undergraduate Research (Bilkent, 2020–2022): developed deep learning methods (transformers, CNNs, deep equilibrium models) for medical imaging, resulting in **2 IEEE Transactions on Medical Imaging papers** and **4 other conference/journal papers** (see [Google Scholar](#)).

SELECTED COURSES AND RELEVANT SKILLS

CMU: Ph.D.-level Optimization and Statistics courses, Generative AI, MDPs and RL, Deep RL

Bilkent: Linear Algebra, Stochastic Models, Signals and Systems, Digital Signal Processing

Skills: Python, PyTorch, Hugging Face Libraries, vLLM, VERL, OpenCV, Apache Spark

PROFESSIONAL SERVICE

- Reviewer for NeurIPS, ICLR, ICML, AISTATS, ISIT, AAAI, TMLR, IEEE/ACM ToN and MLSys
- Teaching Assistant for [Introduction to ML for Engineers \(18-661\)](#) and [Algorithms for Large-scale Distributed ML and Optimization \(18-667\)](#)

SELECTED AWARDS AND ADDITIONAL INFORMATION

- 2023-2024 Ben Cook Presidential Graduate Fellowship at CMU
- 2022-2023 Carnegie Institute of Technology Dean's Fellowship at CMU
- Bilkent University Comprehensive Scholarship: Full tuition waiver & stipend during the B.Sc.
- Turkish Government Merit Scholarship: Awarded stipend during the B.Sc.
- Ranked 252nd/2M in the university and 3rd/104k in the postgraduate entrance exams in Turkey
- Volunteer at Young Guru Academy (an international NGO based in Istanbul) from 2018 to 2020